

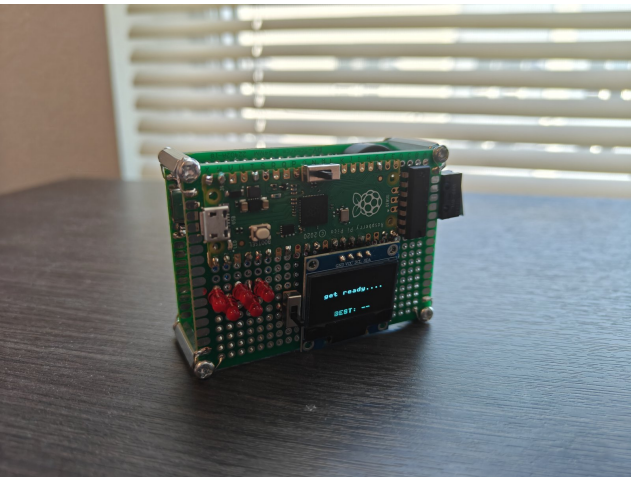
Reaction Time Game

Battery-powered five-light reaction timer with a hardware-debounced input path, checked against an oscilloscope

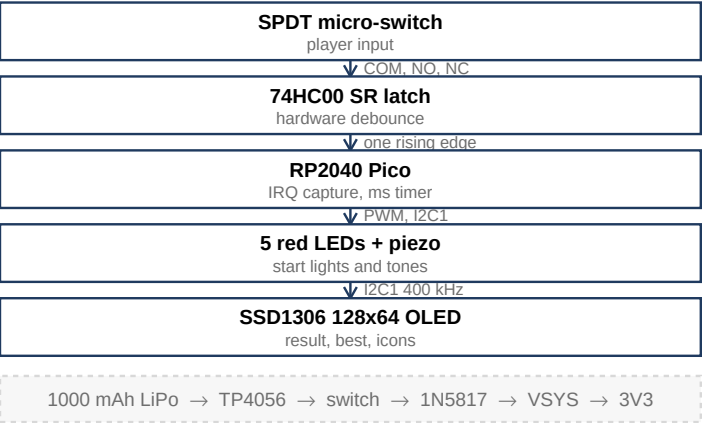
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Full documentation



SYSTEM ARCHITECTURE



WHAT IT DOES

Five LEDs light one second apart, hold, then go out after a random delay. The lights-out instant starts a millisecond timer that one button press stops. A discrete 74HC00 SR latch debounces the button in hardware, so the firmware timestamps one clean edge and contains no debounce code. A 30-round session recorded a median of 202 ms.

HOW IT WORKS

- SPDT micro-switch into a cross-coupled NAND latch: 6 to 11 transitions per press raw, one edge out.
- RP2040 captures that edge on an interrupt armed only during a round; the same handler rejects a false start.
- 1000 mAh LiPo through a protected TP4056 and a Schottky diode-OR onto VSYS; switch measured at 0.2 mV off.

MEASURED RESULTS

Session median, 30 rounds	202 ms
Timer bias vs oscilloscope	+6.4 ms
Contact bounce, raw to latch	6–11 → 1
VSYS with switch off	0.2 mV

KEY DECISIONS

- Hardware SR latch over RC debounce: an RC filter adds 1 to 3 ms to every timestamp.
- Charging with the load switched off keeps termination clean and validates the VSYS gauge.

MEASURED EVIDENCE



Fig. 1 — FNIRSI 2C53T, 0.5 ms/div, 1 V/div, DC: NO opens, NC closes 0.94 ms later



Fig. 2 — FNIRSI 2C53T, 50 ms/div, CH1 20 mV/div AC, CH2 500 mV/div DC, single shot